SMART INDIA HACKATHON 2026

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PRAHARI — The Countdown Band

Warns before the emergency — the flood, cyclone or heatwave coming at you, and the heat strain
building inside you — and reaches workers no tower reaches.



- **Problem Statement ID – 26181**
- Problem Statement Title – A secure, AI-powered Personal Health Companion that delivers real-time, privacy-preserving health monitoring and early warning capabilities, helping individuals recognize health risks before they become emergencies. The solution should improve resilience during heat waves, floods, pollution events, and other disasters common in India while enabling continuous health support through on-device intelligence.
- **Theme – MedTech / BioTech / HealthTech**
- **PS Category – Hardware**
- **Team ID – <your team id>**
- **Team Name – Point Break**

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PROPOSED SOLUTION

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Problem at Hand



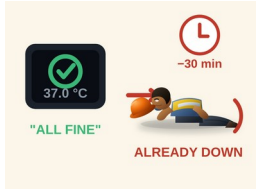
The warning never arrives.

2 in 3 flood-exposed Indians get none.



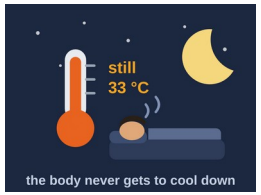
The network dies first.

Kerala 2018: towers ran out of diesel.



The body warns too late.

Collapse is the first symptom noticed.



Nobody is watching at night.

Warm nights rise faster than hot days.

Our Solution — Two Products, One System



1 · THE BAND
Counts down

UNDERSIDE
PPG and IR optics

2 · THE APP
Fetches CAP

A Disaster Warning That Reaches Everyone

IMD · CWC · NDMA
issue the alert

➡

SACHET CAP
any phone with signal

➡

LoRa hop
band to band, ~2 km

➡

Your wrist
buzzes — no phone

Flood · Cyclone · Heatwave · Tsunami · Lightning · Forest fire · Avalanche

Why We Stand Out

- **Warns before, not after —** the hazard outside, the strain inside.
- **Works when the tower doesn't —** the alert hops band to band.
- **Official alerts, not guesses —** the NDMA SACHET CAP feed.
- **₹3,468 of parts —** against ₹28,000 for the rival.

Key Features

Multi-hazard early warning
NDMA SACHET CAP — flood, cyclone, heat

LoRa peer relay
SX1278, 433 MHz, band to band, no tower

Minutes-to-threshold trend
ECTemp from heart rate, RMSE 0.30 °C

Wrist skin temperature
MLX90614 infrared, 0.20 °C MAE

PRAHARI app over BLE
fetches CAP, keeps the offline ledger

Thermal debt ledger
carries strain across the night

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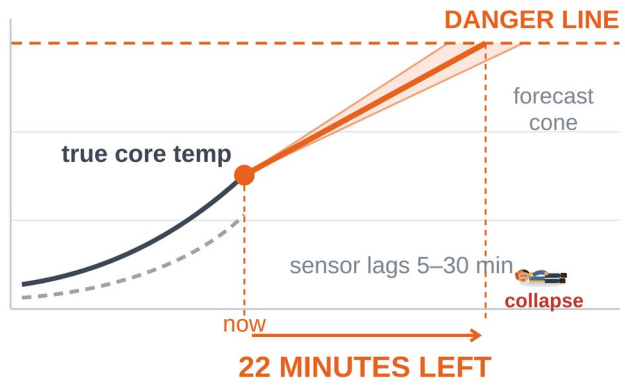
TECHNICAL APPROACH

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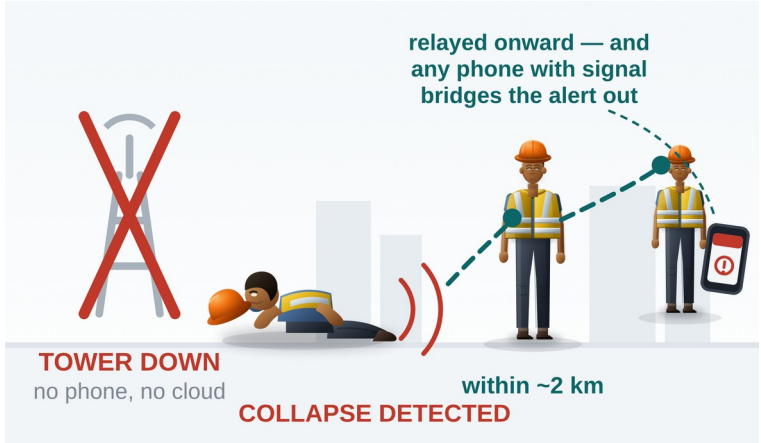
1 · SENSE ON THE BODY

- **XIAO nRF52840 Sense** — Cortex-M4F, IMU and mic already on board
- **MAX30100 red/IR PPG** — heart rate, gated on the IMU when motion is high
- **MLX90614 infrared** — wrist skin temperature, 0.20 °C MAE, no contact
- **AHT20** — air temperature and humidity, the WBGT inputs



2 · DECIDE ON THE CHIP

- **TensorFlow Lite Micro** — ~1 ms inference, ~18 ms with feature extraction
- **ECTemp Kalman + WBGT + Strain Index** — fused every second
- **Welford + CUSUM** — learn this wearer, on-device
- **Forecast to threshold** — gated on a signal-quality index



3 · WARN WITHOUT A NETWORK

- **Haptic warning** — one pattern for hazard, another for your own strain
- **SX1278 LoRa, 433 MHz** — band to band, licence-exempt under G.S.R. 347(E)
- **NDMA SACHET CAP relay** — flood, cyclone and lightning warnings flood the mesh
- **AES + ATECC608A** — only anonymised counts ever sync

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FEASIBILITY AND VIABILITY

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Technical Feasibility



- **Four modules, nothing custom** — all stocked on Robu.in today
- **Every algorithm is published** — ECTemp, Liljegren WBGT, Strain
- **SACHET CAP is a public feed** — rss_india.xml, no key needed
- **IMU and mic already on board** — fewer parts to fail

Economical Feasibility

Parameter	Commercial	PRAHARI
Device cost	₹28,000+	**₹3,468**
Works with no network	No	**Yes**
Warns before impact	No	**Yes**
Output	Lagging °C	**Minutes left**
Mesh rescue alert	None	**~2 km**
Disaster alerts	None	**All CAP hazards**

Sources: greenTEG CORE 2 (mindtecstore.com). PRAHARI = XIAO nRF52840 Sense ₹2,299 + MLX90614 ₹699 + SX1278 Ra-02 ₹329 + MAX30100 ₹85 + AHT20 ₹56 — Robu.in, Aug 2026.

Risks and Mitigation

IR PPG fails under motion	IMU gates it; a bad window raises no alarm, ever
No globe, so no true MRT	We report skin and air temperature, not radiant load
SACHET feed unreachable	Last valid CAP is cached and still relayed, with its age shown
433 MHz limits in India	10 mW e.r.p. / 10% duty — G.S.R. 347(E), 2022

Operational Feasibility

- **The band needs no phone** — a vibration and a falling number
- **The app is optional** — it adds history, not the decision
- **On-board LiPo charger** — BQ25101, charged like a phone
- **Sealed, washable band** — no exposed connectors on the strap

Implementation Viability

Manufacturing	Five off-the-shelf modules, no custom silicon
Availability	Every part in stock on Robu.in today
The app	Flutter, Android first, offline-first storage
Readiness	Bench prototype now; field-testable in 12 weeks

Regulatory and Environmental

- **Class A wellness device** under CDSCO MDR-2017 — we flag, we do not diagnose
- **DPDP Act §7(f) and §7(h)** allow emergency transmission without consent
- **Heatwaves are a notified disaster** since Aug 2026 — SDRF funding applies
- **Licence-exempt radio** — 433.05–434.79 MHz, G.S.R. 347(E); ETA by self-declaration

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IMPACT AND BENEFITS

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30.4 cr

informal workers registered on e-Shram

Ministry of Labour, Dec 2024

1 in 3

flood-exposed Indians have any early warning

CEEW, multi-hazard EWS study

0%

heat-stroke mortality when cooling starts in time

Medicina 2020, n = 521

₹6.5 lakh cr

lost in wages to heat each year — 1.99% of GDP

IIED / Somanathan et al.

Who It Protects



Construction workers



Farm and salt-pan workers



The elderly, living alone



Pregnant women

Social

- **A warning where none exists today** flood, cyclone and heat, to people no tower reaches
- **No smartphone, no literacy needed** — a vibration and a falling number
- **One band protects a whole gang** — a colleague becomes the first responder

Economic

- **₹3,468 against ₹28,000** for the nearest commercial sensor
- **24 lost workdays a year** and ~10% of annual earnings — recoverable
- **Fundable from SDRF** now that heatwaves are a notified disaster

Systemic and Environmental

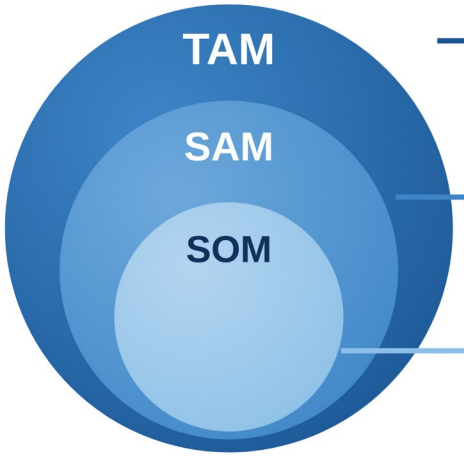
- **Feeds the ASHA NCD app** already on 7.19 lakh tablets, and ABHA
- **Turns IMD's subdivision forecasts** into street-level ground truth
- **2,825 lightning deaths in 2024** India's biggest natural-cause killer — NCRB

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RESEARCH AND REFERENCES

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Market Opportunity



₹1.06 lakh cr
30.4 cr workers × ₹3,500 a unit

₹19,250 cr
5.5 cr heat-exposed outdoor workers

₹193 cr
1% in year one → 5.5 lakh units, one band per gang of ten on a site

Project Links and Documentation

EVERYTHING WE HAVE PUBLISHED

1) Design documentation
2) Firmware repository
3) Sensor test data
4) Bill of materials
5) Demo video
6) Validation report

References

1) Buller et al., USARIEM ECTemp — core body temperature from heart rate; RMSE 0.30 °C. usariem.health.mil
2) "Validity of the CORE Sensor," Sensors, 21(17):5932, 2021 — only 45–51% of readings within 0.3 °C.
3) Filep et al., Medicina, 56(11):589, 2020 — 0% mortality in 521 cases cooled faster than 0.15 °C/min.
4) Moran, Shitzer and Pandolf, Am. J. Physiol., 275:R129, 1998 — the Physiological Strain Index.
5) Lee et al., Sensors, 19(3):673, 2019 — IR PPG carries ~10 dB more motion artefact; hence IMU gating.
6) Liljegren et al., WBGT from standard meteorological measurements, 2008. NWS WBGT guidance.
7) CEEW, Mapping Climate Risks of Extreme Heat in Indian Districts, 2025 — 57% of districts high risk.
8) CEEW — 66% of Indians face extreme floods; only 1 in 3 has any early warning. ceew.in
9) NDMA — satellite backup covers 81 of India's 780+ districts. ndma.gov.in
10) IIED / Somanathan et al. — ₹6.5 lakh crore annual heat wage loss; ~24 workdays lost per worker.
11) DPDP Act 2023, §7(f) and §7(h) · CDSCO/MD/GD/MDSW/01/2026 software guidance.
11b) G.S.R. 347(E), 9 May 2022 — 433.05–434.79 MHz licence-exempt, 10 mW e.r.p., 10% duty.
12) Costing: greenTEG CORE 2 on mindtecstore.com; Robu.in and Robocraze, Aug 2026; LCSC 1k tier.
13) NDMA SACHET public CAP feed — sachet.ndma.gov.in/cap_public_website/rss/rss_india.xml
14) NCRB, Accidental Deaths and Suicides in India 2024 — 2,825 lightning deaths, 35.7% of all.
15) Sensors, 24(6):1944, 2024 — MLX90614 wrist temperature wearable, 0.20 °C MAE.